



Determinants and performance outcomes of artificial intelligence adoption: Evidence from U.S. Hospitals

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ABSTRACT

Integrating Artificial Intelligence (AI) technology in hospitals offers a unique opportunity to improve hospitals' operating and financial performance. This study is among the first to investigate the determinants and subsequent performance outcomes associated with AI adoption. Using an extensive dataset encompassing 941 AI hospital-year observations and 941 non-AI hospital-year observations, we find that hospitals with a larger market share are great candidates to adopt AI. Furthermore, these hospitals can leverage AI technology to enhance various aspects of performance, including total outpatient revenue, total inpatient revenue, productivity, and occupancy. Importantly, we demonstrate that controlling for endogeneity is essential in assessing the performance outcomes of AI adoption. Our findings shed light on the determinants of AI adoption decisions in healthcare and underscore the manifold benefits AI technology brings to hospital operations and financial outcomes.

1. Introduction

Artificial Intelligence (AI) and data analytics are increasingly prevalent in healthcare since these technologies have been approved as effective tools for helping not only patients but also physicians and healthcare staff. These technologies can help patients by reducing waiting times and making healthcare more affordable. They also help physicians to provide more accurate diagnostics and reduce burnout from paperwork and administrative tasks. According to Accenture's (2020) analysis, key clinical health AI applications can potentially create \$150 billion in annual savings for the United States healthcare economy by 2026.

Nuance (2018) reports a case study of its Computer-Assisted Physician Documentation™ (CAPD) at Universal Health Services, Inc. (UHS), a Fortune 500 company with 26 acute care hospitals in the U.S. and over 200 behavioral health facilities in the U.S. and England. UHS has over 70,000 employees and a combined annual revenue of \$10 billion. CAPD, an intuitive cloud-based speech, and Clinical Documentation Improvement (CDI) workflow help UHS to engage physicians in quality improvement. CAPD tackles the challenges of engaging independent physicians in documentation-improvement efforts, reducing transcription costs, increasing documentation speeds, and capturing more

accurate and detailed records of patient conditions to improve quality measures and reimbursement. In particular, this project achieves a 69 % reduction in transcription costs, netting \$3 million in savings annually; gains a 12 % increase in the case mix index (CMI), which is used in determining the allocation of resources to treat patients; improves documentation of patients with extreme severity of illness by 36 %; and expands the details recorded on patients with the highest risk of mortality by 24 %.

While academic and industrial interest in AI applications in healthcare is growing, there is no consensus on whether AI adoption influences hospital performance (Raimo et al., 2022). Recognizing the promising AI benefits, the literature also highlights barriers to AI adoption, including the cost of AI infrastructure and implementation (Kraus et al., 2021), in-house AI expertise (Dicuonzo et al., 2022), and data challenges (Sun & Medaglia, 2019), among others. These factors can diminish returns on hospitals' AI investments, thus reducing the rate of AI adoption. Given the scarcity of rigorous empirical research in this field (Choudhury & Asan, 2020), it is critically important to provide statistical evidence of whether AI adoption improves hospitals' efficiency and effectiveness (Coccia, 2020; Secinaro et al., 2021).

This study aims to fill gaps in the literature by investigating the determinants of AI adoption in U.S. hospitals and the impact of AI adoption

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on hospital financial and operational performance by utilizing extensive sample data. We collect 941 AI hospital-year observations and 941 non-AI hospital-year observations across 40 states and spanning the period of 2000–2020. Drawing upon the principles of the organizational information processing theory (OIPT), we conjecture that market share and average length of stay could be primary determinants to influence the adoption of AI. Notably, a substantial market share equips hospitals with the necessary resources to seamlessly implement AI, thereby bolstering the quality of healthcare delivery. The extended average length of stay is linked to heightened rates of medical complications and increased uncertainty. OIPT posits that hospitals require superior information processing capacity as their tasks involve more uncertainty and intricate interdependence. Hence, hospitals need to enhance their information-processing capacity to effectively address this concern. Furthermore, a substantial market share equips hospitals with the necessary resources to seamlessly implement AI, thereby bolstering the quality of healthcare delivery.

Consistent with the OIPT, we find that hospitals with higher market share have greater incentives to adopt AI to further advance their financial and operational competitiveness. However, we fail to find evidence that hospitals with a higher average length of stay are more likely to adopt AI. The other interesting finding is that AI adoption is significantly positively associated with hospital performance measured by total outpatient revenue, total inpatient revenue, productivity, and occupancy. The benefits of AI adoption might not immediately be reflected on the financial statement due to the learning curve of AI technologies and initial investment, but hospitals will benefit from AI adoption in the long term.

Motivated by the research of Coccia (2020) and Secinaro et al. (2021) on the impact of AI adoption on hospital performance, we employ a large sample of AI and non-AI hospitals and perform a rigorous analysis of what factors can influence AI adoption in hospitals and how AI adoption can impact hospitals' operational and financial performance. Our study is one of the first to use secondary data to investigate these complex relationships; existing literature, in contrast, has largely relied on case studies or surveys to examine the benefits of AI in healthcare (Choudhury & Asan, 2020; Dicuonzo et al., 2022; Leone et al., 2021). Though case studies or surveys can illuminate the successful AI adoption journeys of individual hospitals, they might not offer universal guidelines applicable to all healthcare institutions or address the suitability of AI adoption for different types of hospitals. Our study, however, examines the ramifications of AI adoption, distinguishing itself from other studies that delve into the convergence of AI with other Healthcare 4.0 (H4.0) technologies (Raimo et al., 2022; Tortorella, Fogliatto, Esposto, et al., 2022).

Beyond its contribution to the current body of literature, our study also offers practical guidance to professionals in the field. We examine the influence of AI adoption at large hospitals from various aspects, such as impacts on operations, workload of the healthcare workforce, and operating and financial performance. It is almost impossible for these large hospitals to continue improving their performance without adopting breakthrough AI technology, whereas small hospitals might be able to roll out some traditional, non-AI projects to improve their performance. Our results suggest that hospitals with a higher market share have greater incentives to adopt AI to further advance their financial and operational competitiveness. We hope these findings can help hospitals make a more informed decision when they evaluate AI adoption at their organizations.

The remainder of this article is organized as follows: Section 2 reviews the literature; Section 3 presents the conceptual framework of this study and the hypotheses development; Section 4 presents the details of our research design; the results of our analyses and the findings of our study are discussed in Section 5. Finally, Section 6 concludes with a discussion of the contributions and limitations of this study and suggestions for future research.

2. Literature review

2.1. AI adoption in healthcare

The literature offers several definitions of AI. Some studies view AI as a machine (McCarthy et al., 1955), an agent, or a system (Poole et al., 1998) that can perform as human brains do. Others focus on such technologies as machine learning, deep learning, natural language processing, cognitive computing, neural networks, and similar systems that allow machines to display human intelligence (Agrawal et al., 2018; Liu et al., 2020). Most scholars agree that AI can process, analyze, and interpret data to learn to make decisions independently to achieve objectives and tasks (Kaplan & Haenlein, 2019; Liu et al., 2020; Paschen, Wilson, & Ferreira, 2020).

In healthcare settings, major AI technologies being used include natural language processing, machine learning, and deep learning. Natural language processing is the ability to analyze human language, which enables computers to recognize, interpret, and manipulate human language (Paschen, Pitt, & Kietzmann, 2020). Healthcare systems use natural language-processing techniques to extract relevant information from narrative texts or images to provide data for clinical decision systems. Machine learning refers to a system's ability to continuously improve its capabilities and performance of a specific task by progressively analyzing and acquiring knowledge from data (Murphy, 2012). Deep learning is a subset of machine learning that allows computers to gain knowledge from experience and comprehend the world through a hierarchy of concepts (Goodfellow et al., 2018). Machine learning and deep learning have been used to process patients' medical records to improve diagnoses and clinical decision-making.

The healthcare operations within an organization encompass activities pertaining to finance, administration, and quality enhancement, all aimed at fortifying fundamental treatment functions. Some of the recurring challenges of healthcare services are enhancing patient outcomes and reducing high costs while improving efficiency (Andel et al., 2012; Porter et al., 2021; Weiner et al., 2020). Many healthcare organizations are adopting AI technologies over traditional ones to tackle the above issues thanks to AI's distinct ability to gather and process data to produce a definite output for the end user. While AI technologies, models, and applications in healthcare are developing at an increasing rate and their promising benefits in improving healthcare delivery are evident (Fazal et al., 2018; Hassanzadeh et al., 2019), medical practitioners are slow to adopt them in current healthcare processes (Pan et al., 2019; Smith et al., 2021). Existing literature highlights numerous challenges impeding the adoption of AI in healthcare. Prominent concerns include factors such as culture and personnel, profitability, technology, and data (Fan et al., 2020; Panch et al., 2019; Sun & Medaglia, 2019).

2.2. Determinants of AI adoption in healthcare

Empirical research on AI adoption in healthcare has tried to explore what factors enable users (Alaiad & Zhou, 2014; Saheb, 2020; Shareef et al., 2021) and organizations (Benzidia et al., 2021; Damoah et al., 2021; Leone et al., 2021) to embrace this technology. In terms of users, factors impacting adoption behaviors are different between healthcare professionals (e.g., doctors, nurses, administrators, etc.) and patients. For healthcare professionals, initial trust, performance expectancy, belief, and previous experience in these AI applications (Fan et al., 2020; Pan et al., 2019; Prakash & Das, 2021) are among the factors facilitating favorable opinions of AI. For patients, the major concerns when deciding to use AI technology include personalization, social influence, data risk, and trust (Alaiad & Zhou, 2014; Saheb, 2020; Shareef et al., 2021).

Despite the amount of attention on AI adoption in healthcare at the individual level, empirical research on the factors that shape AI adoption in healthcare organizations is limited (Khanijahani et al., 2022). Some elements, including hospital ownership, years of operation, number of

employees, number of inpatient beds, teaching status, and size have been found to be determinants of AI adoption in hospitals (Raimo et al., 2022; Tortorella et al., 2020). Culture (such as the levels of stakeholders' involvement and organizational attitude) is another factor that facilitates AI integration in hospital operations (Watson et al., 2020). However, there are persistent calls to identify additional determinants influencing AI adoption in healthcare (Bajgain et al., 2023; Khanijahani et al., 2022). Gaining insights into these factors holds paramount importance, as this can not only accelerate AI adoption rates within healthcare but also amplify hospitals' capacity to fully harness the benefits offered by AI.

This study examines market share and average length of stay as determinants of AI adoption. While these factors have been viewed as significant antecedents in healthcare settings (Philbin et al., 2001; Sud et al., 2017), their roles in an AI context remain unexplored. As an important factor for planning and management of care (Pallin et al., 2013), the average length of stay is a significant motivation for AI adoption because a longer length of stay increases readmission risk and mortality rate, accelerating healthcare costs significantly and negatively affecting patient satisfaction (Sud et al., 2017). Naturally, it is a valuable objective for managers of hospitals with longer lengths of stay to try different practices, mechanisms, and technologies, including AI, to reduce this quality indicator (Sarpong et al., 2022). Therefore, examining the effect of length of stay on AI adoption would offer suggestive insights into the issue of longer lengths of stay.

Furthermore, hospital market share is also a critical determinant in healthcare settings. Hospitals with a high market share try to acquire the

attention of stakeholders (e.g., physicians, insurers, patients) through various decisions about new resources and operational strategies. One example of this is hospitals' investment in new technologies like AI. These actions are important to stakeholders because they provide more information about hospitals' underlying capabilities and distinguish certain hospitals from others (Basdeo et al., 2006). However, technologies like AI require significant time and financial resources for new equipment and employee training (Watson et al., 2020). While it is suggested that hospitals with a larger market share would enable AI adoption, they must also effectively attain favorable outcomes and operational efficiency through resource utilization. Consequently, delving into the impact of market share on AI adoption offers valuable insights, guiding the formulation of optimal strategies for implementing AI within these healthcare institutions.

2.3. Performance of AI adoption in healthcare

Empirical research on the impact of AI adoption on hospital performance is exceedingly rare (Choudhury & Asan, 2020; Secinaro et al., 2021; Sun & Medaglia, 2019). Most empirical studies investigate AI in combination with other H4.0 technologies (see Table 1). Aceto et al. (2018) suggest grouping these healthcare technologies into four pillars: Sensing, Communication, Processing (which includes AI), and Actuation. Tortorella, Fogliatto, Esposto, et al. (2022) combine H4.0 technologies into Sensing-Communication and Processing-Actuation bundles and find that they have a positive impact on hospital performance in terms of cost, productivity, patient satisfaction, and patient

Table 1
Summary of Empirical Research on the Impact of AI Adoption on Hospital Performance.

No	Articles	Objectives	Methodologies	Determinant Factors	Performance Indicator	Findings
1	(Tortorella, Fogliatto, Esposto, et al., 2022)	Studies the impact of healthcare 4.0 (H4.0) technologies on hospitals' performance	Survey		Cost Productivity Patient satisfaction Patient safety	H4.0 technologies are associated with the hospitals' performance.
2	(Tortorella et al., 2020)	Assesses the influence of contingency factors on the adoption of H4.0 technologies and barriers to H4.0 adoption	Survey	Hospital ownership Hospital age Number of employees Number of inpatient beds Teaching status		Contingencies do impact H4.0 technologies' adoption and related barriers.
3	(Tortorella, Fogliatto, Kurnia, et al., 2022)	Recognizes bundles of H4.0 digital applications and assesses their impacts on patient-centered performance	Survey		Patient readmission rate Patient mortality rate Patient length of stay Patient satisfaction Patient safety	Hospitals with high H4.0 adoption scores are positively associated with patient-centered performance.
4	(Raimo et al., 2022)	Identifies factors that facilitate digital transformation of hospitals	Survey	Hospital size Hospital age Hospital complexity Teaching status		Size, age, and teaching status of hospitals have a positive impact on digital transformation.
5	(Gu et al., 2019)	Investigates the impact of post-adoption use of case-based health knowledge system (CBHKS) on the long-term performance of hospital management	Survey		Hospital management performance Hospital management maintenance Market performance	The relationship between CBHKS implementation and hospital management performance is mediated by group effectiveness.
6	(Kumar et al., 2021)	Recognizes the components of responsible AI in healthcare and assesses its impacts on value formation and market performance	Survey			Patient's cognitive engagement and instrument value mediate the relationship between responsible AI solutions and market performance.
7	(Tortorella et al., 2021)	Investigates the impact of ten H4.0 digital technologies on four abilities of resilient systems	Survey		Resilience abilities (monitor, anticipate, respond, and learn)	Four H4.0 digital technologies positively influence four resilience abilities.

safety. Using a different approach, Tortorella, Fogliatto, Kurnia, et al. (2022) group these technologies into a smart healthcare supply chain, a smart patient diagnosis, a smart patient treatment, and a smart patient follow-up bundle and find that they improve patient-centered performance indicators such as patient readmission rate, mortality rate, length of stay, satisfaction, and safety. Furthermore, Kumar et al. (2021) investigate responsible AI solutions in healthcare and find that they help hospitals improve market performance. Finally, AI supports hospitals' resilience by improving their monitoring, anticipation, response, and learning abilities (Tortorella et al., 2021).

From a supply chain perspective, Leone et al. (2021) suggest, through a case study, that AI-based solutions co-created by technology service providers and customers in the healthcare ecosystem can facilitate the creation of high-quality service innovation, thus improving patient care performance. In addition, AI technology supports the collaboration among green healthcare supply chain partners, enhancing hospital environmental performance (Benzidia et al., 2021). As a result, AI-driven applications may help the healthcare supply chain to achieve its sustainable development goals (Damoah et al., 2021).

From the existing literature, the effect of AI adoption on hospital performance remains unclear, and many questions need to be answered. First, the impact of AI implementation has usually been studied together with that of other H4.0 technologies as a bundle, making it unclear whether AI independently supports the improvement of hospital performance. Second, most of the related research has collected subjective survey data about AI adoption in a few hospitals at a certain point in time (see Table 1). In the absence of control groups, the change in the performance of AI-adopted hospitals over time remains vague in the literature. Moreover, focusing on only a few hospitals would limit the generalization of the findings. The limitation of extant studies in AI adoption warrants further empirical investigation using longitudinal data with a large sample size. Third, current research has not investigated the effect of adopting AI on hospital financial and operational performance. These indicators are fundamental for hospitals' managers to judge whether their investments in AI have achieved the desired financial and operational performance (Secinaro et al., 2021).

3. Conceptual framework and hypotheses development

3.1. Organizational information processing theory (OIPT)

According to the Organizational Information Processing Theory (OIPT), as an organization expands, it assimilates diverse internal and external processes, thereby inheriting their inherent complexity and unpredictability (Thompson, 1967). This uncertainty stems from factors such as standard operational procedures, interdependence among sub-units, and the dynamic nature of the external environment (Gattiker & Goodhue, 2004). In response to this uncertainty, organizations may choose to enhance their information processing capacity, a strategic investment aimed at proficiently managing information flow (Galbraith, 1974).

Hospitals are a domain that considerably necessitates an augmented information-processing capacity. First, modern hospitals have various interdependent departments, each possessing distinct objectives and priorities. Despite their differences, these departments must collaborate harmoniously to provide an expanding array of services. Second, hospitals face the challenge of addressing the diversity in treatment outcomes due to the varying characteristics, conditions, and severity of illnesses among patients (Kravitz et al., 2004). Third, a high CMI results in greater uncertainty in clinical decision-making, primarily due to the intricacies of diagnosing and treating patients within a complex case mix. Consequently, the provision of accurate and timely patient clinical information becomes imperative to alleviate this uncertainty and enhance the quality of clinical decision-making.

As outlined by the OIPT, organizations, if equipped with solid information-processing capabilities, would be able to predict

environmental changes, operate efficiently, and consequently attain high organizational performance (Galbraith, 1974). In line with the OIPT, AI in this study is conceptualized as a tool that amplifies the information-processing capabilities of hospitals (Kaplan & Haenlein, 2019). Subsequently, the integration of AI into hospitals' operations would serve to alleviate the intricacies and uncertainties they encounter. Equipped with sophisticated techniques and algorithms, AI systems have the capacity to efficiently handle the vast and intricate medical data of patients. This enables hospitals to derive timely and valuable insights that, in turn, enhance the accuracy of diagnoses and the effectiveness of treatments (Deist et al., 2020). Moreover, the OIPT suggests that hospitals can improve patient outcomes by employing information-processing methods that mitigate ambiguity in clinical environments (Flynn & Flynn, 1999).

3.2. Determinants of AI adoption

3.2.1. Market share and AI adoption

As inferred from OIPT, hospitals are urged to invest in bolstering their information-processing capacity to effectively manage the escalating volume and diversity of complexities and uncertainties. In the healthcare industry, sources of uncertainty include the dynamic external environment, patient demands, interconnected departments, and the variability in patient treatments. Hospitals with a larger market share encounter a more extensive patient base and increased demand, amplifying the unpredictability of medical activities. Concurrently, the proliferation of patient data and the imperative to process this data expand exponentially. Functioning within information-intensive, dynamic, and volatile settings, OIPT recommends that these hospitals consistently gather, analyze, and interpret information adeptly to alleviate uncertainties. By embracing AI, a tool that enhances information-processing capability (Kaplan & Haenlein, 2019), hospitals can navigate the intricacies of their operating environment, substantially improving their overall information-processing efficiency.

However, AI adoption calls for significant investment in technologies such as natural language processing, pattern recognition, machine learning, deep learning, cognitive computing, and neural networks, to name a few. To successfully adopt AI across an entire organization, hospitals also need to spend a significant amount of time and money to educate and train employees. For these reasons, not all hospitals can afford to dedicate the necessary resources to AI adoption. Hospitals with a higher market share have broad physician networks and inpatient bases as well as better access to capital and human resources (Cutler & Morton, 2013; Wani et al., 2018). Moreover, hospitals with a higher market share treat more patients than hospitals with a lower market share. Therefore, those hospitals receive a larger portion of prospective reimbursements from Medicare and a larger portion of insurance reimbursements from other third-party coverage (Lee et al., 2017). It is expected that a larger market share will allow hospitals to dedicate sufficient capital and human resources to AI adoption. In addition to better access to capital and human resources, hospitals with a higher market share can be better able to achieve economies of scale. Compared to hospitals with a lower market share, hospitals with a higher market share can spread costs over a larger number of patients and thus lower the fixed costs per patient.

To the extent that hospitals with a larger market share may have more considerable resources to absorb the costs of adopting AI technologies and achieve economies of scale, we expect a positive relationship between market share and AI adoption. Accordingly, we posit the following hypothesis:

H1a: Hospitals with a larger market share are likely to adopt AI.

3.2.2. Average length of stay and AI adoption

The average length of stay is one of the hospital characteristics affecting the possibility of AI adoption. Extended length of stay often arises due to inadequate coordination, insufficient communication

among healthcare providers, and the unavailability of the patient's medical history during the point of care (Bodenheimer, 2008). Consequently, an extended length of stay is linked to elevated occurrences of medical complications, heightened risk of readmissions, and escalated healthcare expenses. Addressing this challenge necessitates improved operational coordination that spans physical resources, services, and patient flow, given the substantial interdependence among various departments within a hospital.

In line with the OIPT literature, it is noteworthy that the benefits derived from information-processing systems like AI are influenced by the extent of interdependence among sub-units within the organization and the magnitude of uncertainty within various situations (Galbraith, 1974; Wybo & Goodhue, 1995). When dealing with low interdependence, simple coordination modes such as standard operating procedures suffice. However, in scenarios marked by high interdependence, the requirement shifts to a common formalized language and comprehensive, accurate information exchange among sub-units (Thompson, 1967).

According to OIPT, hospitals are encouraged to enhance their information-processing capabilities by investing in advanced information systems. This notion leads to the understanding that hospitals with a higher average length of stay can adopt AI solutions, as AI adoption has demonstrated its ability to elevate the quality of healthcare data, thereby enhancing the overall service-delivery quality. For instance, the incorporation of AI's text-to-speech and speech-to-text capabilities enables doctors to process documents with enhanced accuracy. Additionally, the utilization of AI technologies empowers hospitals to predict ambulatory hospital visits with precision (Hadavandi et al., 2012). This predictive capacity equips hospitals to anticipate the required human resources, supplies, and interdepartmental coordination necessary to provide effective patient care.

Furthermore, hospitals confronted with a high average length of stay have the option to implement telemedicine solutions, allowing for earlier patient discharge and enhancing patient management, particularly in remote areas (Armaignac et al., 2018). The integration of artificial intelligence into remote doctor-patient interactions serves to modernize communication, deepen patient relationships, and facilitate real-time decision-making. For instance, Medtronic's personal diabetes assistant, Suga.IQ, serves as a notable example. It assists patients with diabetes in effectively managing their conditions by alerting them when their blood sugar levels reach dangerous levels, prompting immediate corrective actions (Medtronic, 2018). The Deloitte ASSIT technology demonstrates another instance of communication enhancement in patient care. Leveraging natural language processing and speech recognition, this innovation interprets patients' requests, prioritizes them, and forwards them to the appropriate personnel. Impressively, 90 % of surveyed patients affirm the technology's positive impact on staff responsiveness (Deloitte, 2020).

In scenarios where hospitals are grappling with extended decision-making durations, the significance of a dependable solution like AI becomes more pronounced. Natural language-processing techniques and deep learning come into play to extract information from unstructured data sources, including images, clinical notes, medical journals, operative notes, discharge summaries, and diagnostic testing reports. These methods convert this information into coherent and structured data (Davenport & Dreyer, 2018). In contrast to the unstructured data employed in non-AI hospitals, the structured data employed by AI-driven hospitals proves to be highly advantageous for functions such as storage, searchability, and analytics. This, in turn, significantly enhances the efficiency of tasks and ultimately supports the decision-making process of physicians. Therefore, we expect hospitals facing a high average length of stay will be inclined to implement AI technologies. This inclination arises from the observed positive impact of AI on healthcare delivery quality and the decision-making process. Based on this understanding, we put forth the following hypothesis:

H1b: Hospitals with a higher average length of stay are likely to

adopt AI.

3.3. Performance outcomes of AI adoption

According to OIPT, hospitals with strong information-processing capabilities tend to operate with greater efficiency and attain elevated levels of operational performance. The capacity for information processing not only enriches the quality of patient care but also optimizes the management of hospitals' operational aspects through knowledge application and the utilization of suitable technology (Srivastava & Singh, 2021). Through the adoption of AI, hospitals can significantly augment their information-processing capability (Kaplan & Haenlein, 2019). This advancement holds the potential to significantly enhance their overall capability for processing and utilizing information.

AI-assisted systems possess the capability to transform continually expanding patient medical records into comprehensible, manageable, and actionable information. This transformation serves to mitigate healthcare uncertainty, alleviate information ambiguity, and streamline information dissemination across various caregiver departments and healthcare partners. Additionally, the heightened quality of information enhances hospitals' capacity for demand forecasting, ensuring timely, top-notch care. These systems, furthermore, play a pivotal role in fostering integration within hospitals, connecting hospital-patient interactions, facilitating inter-functional collaboration, and establishing seamless hospital-provider connections (Dobrzykowski & Tarafdar, 2015). These integrations collectively contribute to enhancing hospitals' operational efficiency (Yu et al., 2021). Additionally, according to Tortorella, Fogliatto, Esposto, et al. (2022), AI functions as a processing-actuation technology that translates received information into actionable decisions. Their findings underscore that those hospitals extensively adopting such processing-actuation technologies exhibit better operational performance (Tortorella, Fogliatto, Esposto, et al., 2022).

AI technologies also help healthcare organizations adapt to shifts in the environment and swiftly rebound from disruptions. From the perspective of OIPT, the availability of valuable insights regarding changes in markets, patient demands, and requirements is of paramount importance for healthcare organizations. This information is instrumental in the development of novel products or services, including new drugs and treatment protocols. Organizations that offer diverse services and have multiple revenue streams are better equipped to manage both internal and external disruptions, thereby maintaining their desired performance levels. AI's transformative influence on healthcare is evident in the introduction of innovative services like healthcare chatbots, personalized treatments, and post-discharge remote care. These novel service offerings alleviate the strain on constrained resources such as beds, medical professionals, equipment, and more, all while upholding an elevated standard of service during times of disruption. The research by Tortorella et al. (2021) substantiates that AI stands as one of the technologies bolstering hospitals' resilient capabilities. This includes the ability to monitor, forecast, respond to, and learn from disruptions in order to recover quickly from disruptions and sustain operational and financial outcomes.

As explored in Section 3.2, existing research highlights the transformative impact of AI technologies within hospitals. These technologies streamline patient flow, enhance disease detection, mitigate misinterpretation and diagnostic errors, and bolster the efficiency of medical supply chains and healthcare delivery systems (Kraus et al., 2021; Leone et al., 2021; Fan et al., 2020; Topol, 2019; Damoah et al., 2021; Medtronic, 2018). This, in turn, facilitates the optimization of resources, cost management, and effective healthcare treatment (Leone et al., 2021; Yu et al., 2018). Consequently, the adoption of AI is anticipated to yield improvements in both operational performance and financial performance for hospitals. Based on this premise, we propose the following hypothesis:

H2: There is a positive relationship between AI adoption and hospital performance.

Based on the above discussion, we formulate Fig. 1 to illustrate the conceptual model underpinning this study. In this visual representation, Fig. 1 elucidates two important determinants alongside five distinct performance outcomes associated with AI adoption. The underlying design of this conceptual framework is aimed at substantiating the pathway spanning from determinants to the adoption of AI, and subsequently leading to the resultant performance outcomes attributed to AI adoption.

4. Research design

4.1. Sample selection

To compile a preliminary list of hospitals that have implemented AI, we conduct a meticulous manual data collection process. Employing keywords such as “artificial intelligence,” “machine learning,” “cognitive computing,” “deep learning,” and “natural language processing,” combined with the term “adoption,” we perform comprehensive searches across various sources. This methodology aligns with prior research practices (Choudhury & Asan, 2020; Shinnars et al., 2020). Our search entails publicly accessible data repositories, including hospital websites, practitioner journals, academic journals, and annual financial reports. Additionally, we leverage the Factiva database, procured from Dow Jones & Company, Inc., to augment our data collection efforts.

Following the argument advanced by Zhang (2021), we designate a hospital as a non-AI adopter if our comprehensive search across the aforementioned sources yields no indication of AI adoption within the hospital's context. We adopt a rigorous selection process, retaining solely those hospitals that furnished ample information to ascertain their initial AI-adoption years. This procedure yields 246 AI hospitals with specific adoption years. We then create a matching sample of 246 hospitals that have not incorporated AI into their operations.

To ensure methodological rigor, our selection criteria include the stipulation that non-AI hospitals be situated within the same state as their AI counterparts. This is intended to minimize the potential influence of state-level regulations on healthcare operations, as highlighted by Radley et al. (2017). Furthermore, we require that the non-AI hospitals provide the same category of services as the AI hospitals. This step is taken to account for the systematic differences in objectives that often arise among hospitals offering disparate services, as underscored by Eldenburg et al. (2004). In scenarios where multiple non-AI hospitals satisfy our criteria, our preference is to select the one that most closely aligns with the bed count of the corresponding AI hospital, falling within the range of 90–110 %. This approach aims to ensure a balanced and equitable comparison between the two sets of hospitals.

For our empirical analysis, we draw upon secondary data sourced from the Healthcare Cost Report Information System (HCRIS), a dataset that has been curated and refined by the RAND Corporation's Hospital Data repository. To ensure the consistency of our dataset, we exclude hospitals that are not within the scope of the RAND Corporation's Hospital Data repository. Our ultimate dataset comprises 1,882 observations, collectively reflecting a span of 941 AI-adopting hospital-years and an equivalent count of 941 non-AI-adopting hospital-years. These observations include hospitals situated across 40 distinct states; the data covers the time frame spanning from 2000 to 2020.

Table 2 presents our sample distributions. Panel A shows the number of hospitals that initially adopt AI by year. The earliest adoption year in our sample is 2000, and the year with the most adoptions is 2018. Panel B reports characteristics for both AI and non-AI hospitals. We find that approximately 92 % of AI hospitals are system-affiliated, 42 % are teaching hospitals, 82 % are in metro areas, and 67 % are nonprofit. In contrast, 71 % of non-AI hospitals are system-affiliated, 32 % are teaching hospitals, 75 % are in metro areas, and 60 % are nonprofit.

Panel C displays the distribution of hospital-year observations for both AI and non-AI hospitals, year 0 through year + 4. We denote the initial AI adoption year as year 0, the subsequent year as year + 1, and so

on. We analyze data for the 5-year period from year 0 through year + 4. We have the same number of AI and non-AI hospitals each year, and the number of hospitals decreases from year 0 to year + 4, with year 0 having 246 AI/non-AI hospitals and year + 4 having only 90 AI/non-AI hospitals. Panel D shows the distribution of 1,882 hospital-year observations by year. The years with the largest number of observations are 2018 and 2019, followed by 2020.

4.2. Variable measurements

4.2.1. Indicator of AI adoption

We create an indicator variable, *AI*, to denote hospitals that adopted AI in a particular year. The *AI* indicator has the value of 1 if the hospital adopted AI (i.e., AI hospital), and the value of 0 if the hospital did not adopt AI (i.e., non-AI hospital).

4.2.2. Determinants variables

As discussed in section 3.2, we identify two hospital-specific characteristics that are likely to be associated with the decision to adopt AI: market share (*MKTSHR*) and the average length of stay (*ALOS*). We use *MKTSHR* to measure a hospital's market share. It is calculated as the number of patient days for a hospital over the total number of patient days within the market (Lee et al., 2017; Zhang, 2021), which includes all hospitals within a 25 mi (40.23 km) radius (Zhang, 2021). We use *ALOS* to measure a hospital's average length of stay, and it is calculated as the total number of inpatient days divided by the total number of inpatients discharged (Bai, 2013).

4.2.3. Performance variables

Following the literature, we use three financial measures and two operational measures to assess hospital performance. The three financial performance measures are outpatient revenue (*lnOPREV*), inpatient revenue (*lnIPREV*), and return on assets (*ROA*) (Lee & Choi, 2016; Maiga & Jacobs, 2008; Swink & Jacobs, 2012). *lnOPREV* and *lnIPREV* are constructed as the natural logarithms of outpatient revenues and inpatient revenues, respectively. *ROA* is constructed as net income divided by total assets at year-end.

We use two measures of operational performance that are widely used in the literature: productivity (*PRODUC*) and occupancy (*OCC*) (Bai, 2013; Chang, 2013; Ding et al., 2013; Eldenburg & Krishnan, 2003). *PRODUC* is used to measure full-time employees' output and is constructed by dividing the average daily census by the total number of full-time employees. The average daily census is calculated by dividing the total number of patients per year by 365. *OCC* is used to measure the efficiency in the use of licensed beds and is constructed by dividing the average daily census by the total number of licensed beds.

4.2.4. Control variables

Following the studies on healthcare management, we control for a list of factors that have been documented as affecting the decision to adopt AI and performance outcomes. In particular, we control for total expenses (*lnTotalExp*), complexity and severity of patient illness treated (*CMI*), size of the hospital (*Size*), the proportion of Medicaid and Medicare patients (*Medicare%* and *Medicaid%*), teaching status (*Teaching*), location (*Metro*), ownership (*FP* and *NP*), system affiliation (*System*), state regulation variations (*State*), and year fixed effects (*Year*) (Bai et al., 2010; Bai & Krishnan, 2015; Ding, 2014; Eldenburg & Krishnan, 2003, 2008; Shen et al., 2007; Zhang, 2021).

We use *lnTotalExp* to measure a hospital's total expenses; this figure is calculated as the natural logarithm of total expenses (Aletas, 1999). We use *CMI* to represent the service complexity (Grassetti et al., 2005). *CMI* captures the average weight for the different diagnosis-related groups in a hospital. Hospitals with a higher *CMI* provide more complicated treatments and deal with more severe patients compared to hospitals with a lower *CMI*. *CMI* is calculated as the ratio between Medicare Severity Diagnosis Related Groups (MS-DRG) weights and the

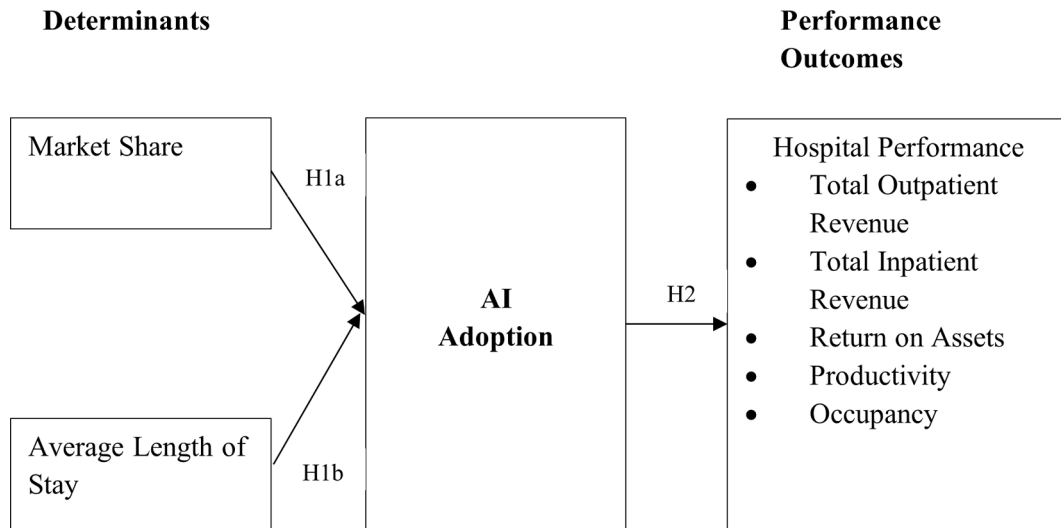


Fig. 1. Conceptual Framework.

total number of discharges, and it is obtained directly from the HCRIS database. *Size* is measured as the natural logarithm of total assets at year-end. *Medicare%* is used to control for the proportion of Medicare patients and is constructed by dividing Medicare patient days by total hospital patient days. *Medicaid%* is used to control for the proportion of Medicaid patients and is constructed by dividing Medicaid patient days by total hospital patient days. *Teaching* is an indicator variable that takes the value of 1 for teaching hospitals and 0 otherwise. *Metro* is an indicator variable that takes the value of 1 for hospitals located in metro areas and 0 otherwise. *FP* and *NP* are used to control for ownership type. *FP* is an indicator variable that takes the value of 1 for for-profit hospitals and 0 otherwise. *NP* is an indicator variable that takes the value of 1 for nonprofit hospitals and 0 otherwise. The government hospital group serves as the reference group in regressions. *System* is used to control for system affiliation. It is an indicator variable that is equal to 1 for system-affiliated hospitals and 0 otherwise. Finally, we use state indicator variables and year indicator variables to control for state and year fixed effects.

4.3. Empirical models

4.3.1. Determinants model

To test our hypotheses H1a and H1b that are related to AI adoption, we estimate logit regressions for the following determinants model framework:

$$AI_{it} = \alpha_0 + \alpha_1 MKTSHR_{it} + \alpha_2 ALOS_{it} + Controls + \varepsilon_{it} \quad (1)$$

In this model, i indexes each hospital, and t indexes time in years. *AI* is an indicator variable indicating a hospital's AI-adoption status. As discussed in H1a and H1b, α_1 , α_2 , and α_3 are expected to be positive. In addition to the determinants, we also include control variables such as *lnTotalExp*, *CMI*, *Medicare%*, *Medicaid%*, *Teaching*, *Metro*, *FP*, *NP*, *System*, and *State* and *Year* indicators.¹

4.3.2. Performance model

To test H2 about the impact of AI adoption on hospital performance, we employ an ordinary least squares (OLS) regression model. AI adoption is expected to be positively associated with a hospital's financial

and operational performance. The performance model is summarized as follows:

$$PERFORMANCE_{it} = \beta_0 + \beta_1 AI_{it} + Controls + \eta_{it} \quad (2)$$

PERFORMANCE is alternatively measured as *lnOPREV*, *lnIPREV*, *ROA*, *PRODUC*, and *OCC*. To be consistent with H2, β_1 is expected to be positive. Control variables include *CMI*, *Size*, *Medicare%*, *Medicaid%*, *Teaching*, *Metro*, *FP*, *NP*, *System*, and *State* and *Year* indicators.

Like existing studies on administrative innovation adoption,² our study is limited by the endogenous nature of AI adoption. That is, the sample of hospitals involved in AI adoption is not random. It is possible that hospitals with better performance are more likely to adopt AI. In that case, there will be a positive correlation between AI adoption and the error term, which leads to an endogeneity problem. This potential endogeneity problem violates the most essential OLS estimation assumption³ (Maiga, 2014), resulting in biased coefficient estimates. To address the potential endogeneity concern regarding hospitals' decision to adopt AI, we follow Adams et al. (2009) in using Wooldridge's (2002) instrumental variable approach, which is a three-stage procedure. In the first stage, we estimate the determinants model—i.e., Equation (1)—and obtain the fitted probability as an instrument for the AI adoption status.⁴ In the second stage, we regress *AI* on the fitted probability and all exogenous variables (i.e., control variables). In the third stage, we regress performance on the fitted value of the second stage and all exogenous variables.

Specifically, the Wooldridge 3SLS regression models are as follows:

$$\begin{aligned} \text{Stage 1: Equation (1)} \\ \text{Stage 2: } AI_{it} &= \mu_0 + \mu_1 \text{FittedProbability}_{it} + Controls + \theta_{it} \\ \text{Stage 3: } PERFORMANCE_{it} &= \delta_0 + \delta_1 AI_{it} + Controls + \zeta_{it} \end{aligned} \quad (3)$$

where *FittedProbability* is the fitted value from the estimation of Equation

¹ To ease the multicollinearity concern, we do not include *Size*, which is highly correlated to *lnTotalExp*. More specifically, *lnTotalExp* is used as a control variable in the determinants model, whereas *Size* is used as a control variable in the performance model.

² Studies include but are not limited to activity-based costing (ABC) adoption (Maiga, 2014), just-in-time (JIT) adoption (Maiga & Jacobs, 2008), and total quality management (TQM) adoption (Westphal et al., 1997).

³ OLS regression assumes the distribution of error terms has zero mean, and there must be no relationship between independent variables and the error terms.

⁴ According to Wooldridge (2002), since the endogenous variable *AI* is a dummy variable, using the fitted probability value from the first stage as an instrument for the *AI* adoption status is better than traditional 2SLS, which uses the fitted probability value from the first stage to replace the endogenous dummy variable (*AI*) in the performance model (Model 2).

Table 2
Sample Distribution.

Panel A: Distribution of Initial AI Adoption Years					
Initial AI Adoption Year		Number of AI Hospitals		Percent (%)	
2000		1		0.41	
2001		1		0.41	
2005		2		0.81	
2006		4		1.63	
2007		2		0.81	
2009		1		0.41	
2011		3		1.22	
2012		9		3.66	
2013		14		5.69	
2014		8		3.25	
2015		10		4.07	
2016		44		17.89	
2017		47		19.11	
2018		90		36.59	
2019		10		4.07	
		246		100	
Panel B: Distribution of Hospital Characteristics					
Hospital Characteristics		Number of AI Hospitals (N=246)	Percent (%)	Number of Non-AI Hospitals (N=246)	Percent (%)
Affiliation	Standalone	21	8.54	72	29.27
	System	225	91.46	174	70.73
Teaching Status	Teaching	103	41.87	79	32.11
	Non-Teaching	143	58.13	167	67.89
Location	Metro	202	82.11	185	75.20
	Urban	41	16.67	57	23.17
	Rural	3	1.22	4	1.63
Ownership	For-profit	60	24.39	61	24.80
	Nonprofit	164	66.67	148	60.16
	Government	22	8.94	39	15.85
Panel C: Distribution of Hospital-Year Observations					
Year	Number of AI Hospitals		Number of Non-AI Hospitals		Total
0	246	246		492	
+1	245	245		490	
+2	211	211		422	
+3	149	149		298	
+4	90	90		180	
Total	941	941		1,882	
Panel D: Distribution of Hospital-Year Observations					
Year	Number of Non-AI Hospitals		Number of AI Hospitals		Total
2000	1	1		2	
2001	2	2		4	
2002	2	2		4	
2003	2	2		4	
2004	2	2		4	
2005	3	3		6	
2006	6	6		12	
2007	10	10		20	
2008	10	10		20	
2009	11	11		22	
2010	9	9		18	
2011	6	6		12	
2012	13	13		26	
2013	27	27		54	
2014	34	34		68	
2015	46	46		92	
2016	89	89		178	
2017	128	128		256	
2018	202	202		404	
2019	202	202		404	
2020	136	136		272	
Total	941	941		1,882	

(1) and *AIFIT* is the fitted value from the second stage. Control variables are the same as those used in Equation (2).

5. Empirical results

5.1. Descriptive statistics

Table 3 reports descriptive statistics for all our variables. Panel A presents the descriptive statistics for the entire sample, whereas Panel B presents the difference in means between AI and non-AI hospitals. Panel A shows that, on average, our sample hospitals have a market share of 28.2 % and incur total expenses of \$319 million. Patients spend an average of 2.79 days in these hospitals. As for the hospitals' financial performance, the total outpatient revenue (*OPREV*) and total inpatient revenue (*IPREV*) have a mean of \$581 million and \$120 million, respectively. The average *ROA* is 5.3 %. Related to operational performance, productivity (*PRODUC*) and occupancy (*OCC*) have a mean of 9.6 % and 54.1 %, respectively. On average, Medicare inpatient days account for 36.2 % of the total hospital inpatient days, whereas Medicaid inpatient days only account for 8.2 %. The median hospital is characterized as non-teaching, nonprofit, in a metro area, and system-affiliated.

In Panel B, the t-tests show that the mean differences in the determinant variables of *MKTSHR* and *ALOS* between the AI and non-AI hospitals are statistically significant. The t-tests also indicate that the two groups of hospitals differ significantly in all performance measures except for *ROA* and all the control variables except for *Medicaid%* and *FP*.

Table 4 provides Pearson correlations for determinants, performance measures, and control variables. As expected, *MKTSHR* is significantly positively correlated with AI adoption. In contrast to our expectation, *ALOS* is negatively correlated with AI adoption. As for performance measures, *lnOPREV*, *lnIPREV*, *PRODUC*, and *OCC* are significantly positively correlated with AI adoption as anticipated, whereas there is no significant correlation between *ROA* and AI. Most of the control variables exhibit significant correlations with AI and performance measures, highlighting the importance of incorporating these factors in our regression models. Since some correlation coefficients exceed 0.5, we test for multicollinearity by calculating the variance inflation factor (VIF). We find that all VIF factors are well below the tolerance value of 10, indicating that the regression analyses do not suffer from multicollinearity.

5.2. Determinants results

Table 5 presents the estimation results of Equation (1), our determinants model. Columns 1–2 examine the effect of each determinant individually while Column 3 considers all three determinants together. Consistent with our expectation, *MKTSHR* is significantly positively related to AI adoption, with a coefficient of 0.474 (S.E. = 0.223) in column 1. This suggests that hospitals with higher market share are more likely to adopt AI, which supports H1a. However, *ALOS* is negatively related to AI adoption, with a coefficient of −0.213 (S.E. = 0.064), which is not consistent with H1b. The results remained similar in the pulled regression shown in Column 3, indicating that multicollinearity is not a concern and the effect of each determinant is not affected by that of the others.

Next, we calculate odd ratios for the logit regression in Column 4. The odds ratio of 1.666 for *MKTSHR* shows that, all else equal, for every one unit increase in market share, we expect to see about a 66.6 % increase in the odds of adopting AI. The odds ratio for *ALOS* is 0.843, indicating that each unit increase in *ALOS* is associated with a 15.7 % decrease in the odds of adopting AI.

Taken together, our findings from Table 5 suggest that hospitals with higher market share are more likely to adopt AI.

Table 3
Descriptive Statistics.

Panel A: Full Sample								
Variable	N	Mean	Median	Std. Dev.	Q1	Q3	Min	Max
Determinants								
MKTSHR	1,659	0.282	0.124	0.325	0.035	0.432	0.000	1
ALOS	1,740	2.790	2.740	0.945	2.406	3.090	0.290	14.843
Performance Measures								
OPREV (\$M) †	1,852	581	351	830	143	715	5.686	9,740
IPREV (\$M) †	1,855	120	42.6	277	11.2	131	0.013	8,020
ROA	1,857	0.053	0.046	0.574	−0.010	0.123	−17.650	11.452
PRODUC	1,873	0.096	0.091	0.047	0.064	0.121	0.002	0.707
OCC	1,876	0.541	0.563	0.214	0.368	0.715	0.004	1.046
Control Variables								
TotalExp (\$M) †	1,879	319	156	530	64.4	350	5.306	7,240
CMI	1,740	1.631	1.608	0.347	1.398	1.816	0.899	4.285
TotalAssets (\$M) †	1,845	433	147	968	5.32	377	0.800	16,400
Medicare%	1,873	0.362	0.351	0.131	0.269	0.443	0.026	0.845
Medicaid%	1,882	0.082	0.057	0.087	0.027	0.106	0	1
Teaching	1,882	0.365	0	0.482	0	1	0	1
Metro	1,882	0.778	1	0.415	1	1	0	1
FP	1,882	0.251	0	0.434	0	1	0	1
NP	1,882	0.623	1	0.485	0	1	0	1
System	1,882	0.803	1	0.398	1	1	0	1

Panel B: Comparison of Non-AI Hospitals and AI Hospitals				
Variable	Non-AI Hospitals Mean (1)	AI Hospitals Mean (2)	Difference (1) – (2)	t-statistic (Difference in means)
Determinants				
MKTSHR	0.253	0.312	0.059	−3.686***
ALOS	2.855	2.725	0.130	2.860***
Performance Measures				
OPREV(\$M) †	411	746	−335	−8.882***
IPREV(\$M) †	84.2	154	−69.8	−5.493***
ROA	0.038	0.067	−0.029	−1.091
PRODUC	0.093	0.098	−0.005	−2.387**
OCC	0.504	0.579	−0.075	−7.768***
Control Variables				
TotalExp (\$M) †	207	431	−224	−9.411***
CMI	1.602	1.661	−0.059	−3.594***
TotalAssets(\$M) †	249	612	−363	−8.188***
Medicare%	0.372	0.352	0.020	3.261***
Medicaid%	0.081	0.084	−0.003	−0.929
Teaching	0.300	0.430	0.130	−5.940***
Metro	0.747	0.810	−0.063	−3.282***
FP	0.252	0.252	0.000	0.000
NP	0.596	0.649	−0.053	−2.380**
System	0.692	0.914	−0.222	−12.604***

This panel presents descriptive statistics for the variables employed in this study. See Appendix A for variable definitions.

This panel presents the differences in means between AI hospitals and non-AI hospitals for the variables employed in this study. *, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

† Descriptive statistics are reported for raw values for ease of interpretation, but our regression models use the natural logarithm of these variables.

† Descriptive statistics are reported for raw values for ease of interpretation, but our regression models use the natural logarithm of these variables.

5.3. Performance results

Table 6 presents the regression results of Equation (2), our performance model. The OLS regression results in Panel A show that AI adoption has a positive effect on outpatient revenue, no significant effect on inpatient revenue, return on assets, and occupancy, and a negative effect on productivity. In particular, *AI* has a coefficient of 0.086 (S.E. = 0.027) in Column 1, where *lnOPREV* serves as the performance measure, indicating a significant positive association between AI adoption and outpatient revenue.

Panel B presents the Stage 3 estimation results of Wooldridge's instrumental variable approach. The results show a significantly positive association between AI adoption and all performance measures except *ROA*. For example, the coefficient of *AIFIT* is 5.886 (S.E. = 1.353), 7.544 (S.E. = 1.856), 0.079 (S.E. = 0.026), and 0.524 (S.E. = 0.145) when *lnOPREV*, *lnIPREV*, *PRODUC*, and *OCC* serve as the dependent variables, respectively. This indicates that AI adoption has a positive effect on outpatient revenue, inpatient revenue, productivity, and occupancy.

The coefficient of *AIFIT* is not significant when *ROA* serves as the performance measure. Thus, H2 is supported in terms of *lnOPREV*, *lnIPREV*, *PRODUC*, and *OCC* but not *ROA*. Our results are also consistent with the findings of existing studies that AI adoption in the healthcare system leads to superior performance (Yu et al., 2018; Leone et al., 2021).

Following Maiga (2014), we assess the relevance of the instrumental variable. A rule of thumb is that the F -statistic for the instrument relevance test should be larger than 10 to ensure that the maximum bias in instrumental variable estimator is less than 10 % (Stock & Staiger, 1997). F -statistics reported in Panel B are all greater than 10, with values of 18.349, 18.294, 17.761, 18.149, and 17.761 for performance measures of *lnOPREV*, *lnIPREV*, *ROA*, *PRODUC*, and *OCC*, respectively. Thus, we claim that our instrumental variable passes the relevance test. Also, we conduct Hausman F -tests to assess endogeneity issues. The Hausman F -tests are all significant, with F -statistics of 842.022, 240.336, 23.856, and 58.978 for *lnOPREV*, *lnIPREV*, *PRODUC*, and *OCC*, respectively. The results indicate that the decision for AI adoption is endogenous, and that the OLS regression will yield biased results. It is worth noting that the

Table 4
Pearson Correlations.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
(1) AI										
(2) MKTSHR	0.090***									
(3) ALOS	−0.068***	0.088***								
(4) lnOPREV	0.238***	0.178***	−0.224***							
(5) lnIPREV	0.205***	0.123***	−0.043*	0.860***						
(6) ROA	0.025	0.035	−0.032	0.094***	0.101***					
(7) PRODUC	0.055**	−0.050**	0.227***	0.318***	0.451***	0.094***				
(8) OCC	0.177***	0.093***	0.045*	0.661***	0.755***	0.091***	0.447***			
(9) lnTotalExp	0.231***	0.184***	−0.154***	0.912***	0.883***	0.054**	0.250***	0.732***		
(10) CMI	0.086***	0.027	−0.517***	0.488***	0.379***	0.005	0.013	0.321***	0.516***	
(11) Size	0.191***	0.190***	−0.069***	0.806***	0.780***	0.120***	0.185***	0.685***	0.907***	0.420***
(12) Medicare%	−0.075***	0.160***	0.130***	−0.392***	−0.359***	−0.023	−0.241***	−0.216***	−0.341***	−0.283***
(13) Medicaid%	0.021	0.055**	0.121***	0.138***	0.196***	0.014	0.141***	0.107***	0.163***	−0.061**
(14) Teaching	0.136***	0.138***	−0.014	0.533***	0.539***	0.023	0.144***	0.473***	0.606***	0.332***
(15) Metro	0.076***	−0.223***	−0.143***	0.476***	0.491***	0.013	0.295***	0.375***	0.488***	0.397***
(16) FP	0.000	−0.194***	−0.180***	−0.073***	−0.091***	0.029	0.223***	−0.153***	−0.197***	0.103***
(17) NP	0.055**	0.082***	0.060**	0.115***	0.153***	−0.005	−0.067***	0.230***	0.197***	−0.001
(18) System	0.279***	−0.072***	−0.088***	0.284***	0.275***	0.040*	0.259***	0.267***	0.223***	0.165***
Variable	(11)	(12)	(13)	(14)	(15)	(16)	(17)			
(11) Size										
(12) Medicare%	−0.244***									
(13) Medicaid%	0.148***	−0.159***								
(14) Teaching	0.522***	−0.245***	0.133***							
(15) Metro	0.424***	−0.388***	0.014	0.296***						
(16) FP	−0.324***	−0.161***	0.011	−0.051**	0.094***					
(17) NP	0.274***	0.080***	−0.140***	0.076***	0.012	−0.746***				
(18) System	0.186***	−0.233***	0.021	0.121***	0.237***	0.118***	0.149***			

This table presents Pearson correlations among the variables employed in this study. See Appendix A for variable definitions.

*, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

Hausman test for ROA is not significant, suggesting that there is no difference between OLS and 2SLS estimates when using ROA as the dependent variable.

5.4. Difference-in-Difference test

To further mitigate endogeneity concerns, we employ a difference-in-difference test to identify the causality between AI adoption and performance.

The difference-in-difference approach is based on a parallel trend assumption, which requires that the difference between the treatment and control group is constant over time if there is no treatment. We use the approach outlined in Anand et al. (2021) to examine whether hospital performance between AI and non-AI hospitals exhibits a parallel trend prior to AI adoption. For each sample year, we identify all hospitals that adopted AI in that particular year. We designate the initial year of AI adoption as year 0 and the subsequent (previous) year as year + 1 (−1). Then we gather data for the periods from year −3 to year + 3 and create a trend variable *Pre-AI Period* to represent the periods before AI adoption. Specifically, *Pre-AI Period* takes the value of 1 for the periods of year −3 to year −1 and 0 otherwise. The interaction term *Pre-AI Period* × *AI*, is used to compare the trend in performance between AI and non-AI hospitals for the pre-AI period. We take our performance measures as the dependent variable and include all the control variables as in Equation (2). The estimation results show that the coefficients of the interaction term are all nonsignificant, which satisfies the parallel trend assumption.⁵

To conduct the difference-in-difference test, we start with a univariate analysis. We first compute the change in performance measures for the treatment and control groups over three periods: from year −1 to year + 1, year −1 to year + 2, and year −1 to year + 3. Our treatment group includes all AI hospitals that we identified in each sample year, while the control group includes all non-AI hospitals. Then we compute

the differences in performance changes between the two groups. The univariate results are reported in Panel A of Table 7. The t -test results illustrate that the differences in all performance changes except the change in ROA are positive and significantly different from zero, indicating that hospital performance improves after AI adoption.

Next, we perform a multivariate analysis to control for possible changes in hospital characteristics owing to changes in AI adoption. Dependent variables are changes in performance measures ($D_{\lnOPREV}$, $D_{\lnIPREV}$, D_{ROA} , D_{PRODUC} , and D_{OCC}). The test variable is AI adoption (*AI*). We control for changes in all control variables in Equation (2). Year and state fixed effects are included. The results are shown in Panel B of Table 7. Consistent with the univariate results, the coefficient estimates of *AI* are significantly positive across all the columns except for the D_{ROA} column. The findings suggest that, in comparison with the matching sample, performance in terms of outpatient revenue, inpatient revenue, productivity, and occupancy increases more for hospitals with AI adoption.

Lastly, we use graphs to illustrate the dynamics of the effect of AI adoption on hospital performance. Fig. 2 presents the evolution of performance in both the treatment (dashed line) and control groups (solid line) over a span of six years, with two years before and four years after the AI adoption.⁶ Five performance measures, *lnOPREV*, *lnIPREV*, *ROA*, *PRODUC*, and *OCC*, are displayed in Panels (a) through (e), respectively.

As shown in the figure, the performance of AI and non-AI hospitals exhibit similar trends prior to the initial adoption year (year 0), which confirms the validity of our parallel trends test, i.e., that the treatment and control groups have parallel trends in the pre-event period. After the adoption year, the performance of the two groups largely moves together in the following two years. Then in year + 3 and year + 4, AI hospitals tend to show better performance than their non-AI counterparts. The only exception is ROA, which roughly stays constant for both

⁶ To provide a more comprehensive view of the trends, we have extended the time frame used in our aforementioned difference-in-difference tests from four years (−1 to +3) to six years (−2 to +4).

⁵ The results are available upon request.

Table 5
Determinants of AI Adoption: Logit Model.

Variable	(1)	(2)	(3)
Intercept	-11.645*** (1.624)	-12.601 (1.927)	-11.016*** (1.675)
MKTSHR _{it}	0.474** (0.223)		0.511** (2.27) OR: 1.666
ALOS _{it}		-0.261*** (0.083)	-0.171* (0.092) OR: 0.843
lnTotalExp _{it}	0.569*** (0.092)	0.708*** (0.085)	0.581*** (0.092) OR: 1.787
CMI _{it}	-0.567** (0.239)	-1.150*** (0.262)	-0.816*** (0.270)
Medicare% _{it}	-0.013 (0.610)	-0.024 (0.586)	-0.019 (0.617)
Medicaid% _{it}	-1.367 (0.953)	-1.248 (0.830)	-1.253 (0.969)
Teaching _{it}	0.043*** (0.148)	0.055*** (0.140)	0.078 (0.149)
Metro _{it}	0.086** (0.187)	-0.136 (0.173)	0.090 (0.188)
FP _{it}	-0.187** (0.244)	0.163 (0.228)	-0.233 (0.247)
NP _{it}	-0.136 (0.953)	-0.131 (0.209)	-0.174 (0.226)
System _{it}	1.526*** (0.191)	1.586*** (0.179)	1.528*** (0.190)
State	Included	Included	Included
Year	Included	Included	Included
Pseudo R ²	0.107	0.107	0.109
Chi-Square	0.000	0.000	0.000
[Prob > Chi-Square]			
Number of observations	1,508	1,731	1,508

This table presents the estimation results of Equation (1), our determinants model. The dependent variable is AI, an indicator variable for a hospital's AI adoption status. Sample sizes vary across models because some hospitals do not have data for all performance categories. Robust standard errors are reported in parentheses. OR stands for odds ratio. See Appendix A for variable definitions. *, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

AI and non-AI hospitals over the entire time span. This is consistent with our finding in Table 7 that ROA does not show significant improvement for hospitals that have adopted AI. Taken together, the findings depicted in Fig. 2 further support and validate the results obtained from our difference-in-difference analysis.

5.5. Additional tests

In addition to Wooldridge's instrumental variable approach, we also employ the traditional two-stage least square (2SLS) analysis to test the robustness of our results. The fitted probability from the estimation of Equation (1) is used to replace the endogenous indicator variable, AI, in Equation (2). The results in Table 8 show that the coefficients of *FittedProbability* using the 2SLS analysis are similar to those of *AIFIT* as reported in Panel B of Table 6, which provides further support to our core finding that AI adoption is positively related to hospital performance except for ROA.

It is possible that the effect of AI adoption on hospital performance might be delayed because hospitals need time to learn and make effective use of AI. To account for this issue, we investigate the effect of AI adoption on future performance. Specifically, we replace *PERFORMANCE_{it}* in Equation (3) with *PERFORMANCE_{it+1}*. The untabulated results show that four performance measures (i.e., *lnOPREV_{it+1}*, *lnOPREV_{it+1}*, *PRODUC_{it+1}*, and *OCC_{it+1}*) are still significantly positively related to AI adoption.

To mitigate the impact of outliers on regression results, we excluded observations with performance measures of more than three and less

than negative three z-scores from the sample. Alternatively, we winsorize all continuous variables at the 1st and 99th percentiles. The regression results (untabulated) are similar to those reported in Panel B of Table 6, which confirms that our regression results are not driven by outliers.

6. Discussion and future research

6.1. Discussion and conclusions

Employing a dataset of 1,882 observations spanning the years 2000 to 2020 and guided by the theoretical framework of the OIPT, this study undertakes a comprehensive examination. It commences by scrutinizing the factors that influence hospitals' decisions to adopt AI technology. Subsequently, the study delves into the performance outcomes that stem from the adoption of AI within these hospitals. To address the potential endogeneity concern regarding hospitals' decision to adopt AI, we investigate the impact of endogeneity on the relation between AI adoption and hospital performance using Wooldridge's instrumental variable approach.

Our findings indicate a noteworthy trend: hospitals endowed with a substantial market share are notably more predisposed to embracing AI adoption, in contrast to their counterparts with lower market shares. Thus, our study enriches literature on the determinants of healthcare information technology adoption. Furthermore, our study substantiates that the integration of AI technologies correlates with higher financial and operational performance.

Our results point to the compelling suggestion that hospitals endowed with a larger market share should be strongly encouraged to adopt AI technology. These hospitals, given their size and complexity, operate within a dynamic environment, catering to a diverse array of patients and services. Hence, the benefits they stand to gain from AI technology are significant. By utilizing AI, they can streamline data derived from diverse hardware and software systems, alleviate administrative burdens on physicians, optimize the balance between health resources and patient demand, curtail healthcare costs, and ultimately bolster healthcare efficiency.

A surprising initial outcome emerged when our H1b failed to gain support, indicating that hospitals with higher average lengths of stay are not inherently more inclined to adopt AI. Our interpretation of this outcome revolves around a multifaceted perspective. We postulate that both AI-adopting and non-AI-adopting hospitals share a common aspiration to reduce their average lengths of stay, albeit through divergent means. While non-AI hospitals may opt for other technologies such as lean thinking to achieve this reduction, the overarching objective remains consistent. This convergence in goals could potentially dilute the association between the average length of stay and AI adoption, accounting for the absence of significant findings. Another plausible explanation revolves around the prioritization of AI technologies. In many instances, the central impetus for AI implementation lies in enhancing detection and diagnostic accuracy across both inpatient and outpatient care, rather than focusing exclusively on curbing the average length of stay for inpatients. As long as AI serves to augment physicians' diagnostic accuracy, there exists a compelling rationale for encouraging the adoption of AI across both AI and non-AI hospitals.

The proliferation of advanced AI technologies further accentuates this point. By harnessing AI capabilities, hospitals can optimize operational efficiency and bolster profit margins. This is achieved by utilizing existing hardware, software, and healthcare workforce capacities to accommodate more patients, intensifying system utilization, and reducing patient turnover. This dynamic shift emphasizes the potential for both AI and non-AI hospitals to gain from AI adoption, albeit through different avenues.

Our findings unveil a significant and positive correlation between AI adoption and hospital performance, except in the case of ROA. One plausible rationale revolves around the considerable initial investment

Table 6
Effect of AI Adoption on Performance.

Panel A: OLS Regressions					
Variable	lnOPREV _{it}	lnIPREV _{it}	ROA _{it}	PRODUC _{it}	OCC _{it}
Intercept	7.376*** (0.291)	2.265*** (0.520)	−1.659* (0.911)	−0.074*** (0.019)	−0.967*** (−7.16)
AI _{it}	0.086*** (0.027)	0.056 (0.045)	−0.030 (0.018)	−0.004** (0.002)	−0.001 (0.007)
CMI _{it}	0.138*** (0.058)	−0.160 (0.113)	−0.168 (0.157)	−0.030*** (0.003)	−0.008 (0.016)
Size _{it}	0.537*** (0.014)	0.730*** (0.025)	0.103* (0.054)	0.008*** (0.001)	0.080*** (0.004)
Medicare% _{it}	−0.331** (0.141)	−0.550** (0.270)	0.029 (0.208)	−0.026*** (0.010)	−0.079* (0.045)
Medicaid% _{it}	0.467** (0.197)	2.445*** (0.332)	−0.075 (0.139)	0.042*** (0.015)	0.179*** (0.057)
Teaching _t	0.309*** (0.031)	0.490*** (0.061)	−0.027 (0.024)	0.007*** (0.002)	0.071*** (0.009)
Metro _t	0.247*** (0.041)	0.487*** (0.099)	−0.004 (0.040)	0.017*** (0.003)	0.038*** (0.012)
FP _t	0.342*** (0.054)	0.509*** (0.096)	0.218*** (0.076)	0.037*** (0.004)	0.029** (0.015)
NP _t	0.194*** (0.045)	0.375*** (0.081)	0.052 (0.045)	0.014** (0.003)	0.044*** (0.012)
System _t	0.214*** (0.036)	0.384*** (0.070)	0.015 (0.037)	0.019*** (0.002)	0.053*** (0.010)
State	Included	Included	Included	Included	Included
Year	Included	Included	Included	Included	Included
R ²	0.807	0.729	0.068	0.445	0.607
Number of observations	1,701	1,702	1,704	1,702	1,704
Panel B: Wooldridge's Instrumental Variable Approach					
Variable	lnOPREV _{it}	lnIPREV _{it}	ROA _{it}	PRODUC _{it}	OCC _{it}
Intercept	15.543*** (2.288)	11.403*** (3.145)	−2.290* (1.154)	0.033 (0.044)	−0.534** (0.245)
AIFIT _{it}	5.886*** (1.353)	7.544*** (1.856)	−0.321 (0.296)	0.079*** (0.026)	0.524*** (0.145)
CMI _{it}	0.511* (0.262)	0.341 (0.342)	−0.224 (0.178)	−0.024*** (0.005)	0.023 (0.028)
Size _{it}	0.084 (0.132)	0.146 (0.81)	0.144** (0.073)	0.002 (0.003)	0.041*** (0.014)
Medicare% _{it}	0.061 (0.802)	0.139 (1.014)	−0.074 (0.208)	−0.021 (0.015)	−0.014 (−0.084)
Medicaid% _{it}	−0.003 (1.274)	1.871 (1.633)	−0.089 (0.163)	0.031 (0.024)	0.084 (0.127)
Teaching _t	0.038 (0.198)	0.127 (0.261)	−0.028 (0.023)	0.003 (0.004)	0.045** (0.020)
Metro _t	0.200* (0.224)	0.451 (0.294)	−0.020 (0.046)	0.016*** (0.004)	0.033 (0.024)
FP _t	0.490** (0.269)	1.116*** (0.417)	0.209*** (0.079)	0.041*** (0.006)	0.056* (1.74)
NP _t	0.862* (0.322)	0.691* (0.355)	0.051 (0.055)	0.016*** (0.005)	0.054* (0.026)
System _t	−1.750*** (0.503)	−2.118*** (0.683)	0.105 (0.118)	−0.008 (0.010)	−0.113** (−2.11)
State	Included	Included	Included	Included	Included
Year	Included	Included	Included	Included	Included
Wald chi2	285.30***	354.61***	176.11***	632.31***	806.69***
Instrument relevance F-test	18.349***	18.294***	17.761***	18.149***	17.761***
Endogeneity of AI	842.022***	240.336***	0.937	23.856***	58.978***
Hausman F-test					
Number of observations	1,480	1,482	1,483	1,481	1,483

Panel B presents the Stage 3 estimation results of Wooldridge's instrumental variable approach. Sample sizes vary across models because some hospitals do not have data for all performance categories. Robust standard errors are reported in parentheses. See Appendix A for variable definitions.

*, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

This table examines the effect of AI adoption on hospital performance. Panel A presents the regression results of Equation (2), our performance model. Sample sizes vary across models because some hospitals do not have data for all performance categories. Robust standard errors are reported in parentheses. See Appendix A for variable definitions.

*, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

and the ongoing support demanded by AI adoption. This transformative endeavor necessitates active involvement from physicians, staff, and management teams. The process of fully embedding AI adoption into the organizational culture may span several years. Consequently, the impact of AI adoption on ROA might not manifest significantly in the short term.

While the financial gains stemming from AI adoption may not be immediately evident due to the learning curve associated with AI technologies and the initial investment, the hospitals that commit to AI adoption can anticipate reaping exponential benefits in subsequent years, with the financial returns potentially experiencing a noteworthy

Table 7
Changes in Performance and AI Adoption.

Panel A: Univariate Analysis										
Year-to-Year	Average Log of Outpatient Revenue Change		Difference (1) – (2)	t-Test of (1) – (2)	N	Average Log of Inpatient Revenue Change		Difference (1) – (2)	t-Test of (1) – (2)	N
	Event Sample (1)	Matching Sample (2)				Event Sample (1)	Matching Sample (2)			
–1 to + 1	0.178	0.132	0.046	2.383**	515	0.131	0.056	0.075	2.297**	517
–1 to + 2	0.242	0.188	0.054	2.198**	456	0.188	0.102	0.084	2.272**	457
–1 to + 3	0.301	0.229	0.072	1.912*	304	0.201	0.086	0.115	2.010**	305
Year-to-Year	Average ROA Change		Difference (1) – (2)	t-Test of (1) – (2)	N	Average Productivity Change		Difference (1) – (2)	t-Test of (1) – (2)	N
	Event Sample (1)	Matching Sample (2)				Event Sample (1)	Matching Sample (2)			
–1 to + 1	–0.001	0.015	–0.016	–0.329	518	–0.002	–0.003	0.001	0.860	522
–1 to + 2	–0.070	0.012	–0.083	0.632	458	–0.001	–0.003	0.002	1.944*	461
–1 to + 3	0.026	–0.037	0.063	0.438	306	–0.002	–0.008	0.006	2.494**	307
Year-to-Year	Average Occupancy Change		Difference (1) – (2)	t-Test of (1) – (2)	N					
	Event Sample (1)	Matching Sample (2)								
–1 to + 1	–0.001	–0.017	0.016	2.245**	525					
–1 to + 2	0.003	–0.017	0.020	2.348**	463					
–1 to + 3	–0.007	–0.033	0.026	2.225**	309					
Panel B: Multivariate Analysis										
Periods: –1 to +1										
Variable	D_InOPREV		D_InIPREV			D_ROA		D_PRODUC		D_OCC
AI_{it}	0.063*** (0.021)		0.080** (0.031)			0.025 (0.051)		0.002 (0.002)		0.018** (0.007)
Control variables	Yes		Yes			Yes		Yes		Yes
Year and State	Yes		Yes			Yes		Yes		Yes
Fixed effects										
R^2	0.153		0.178			0.218		0.126		0.143
Number of observations	474		475			477		475		477
Periods: –1 to +2										
Variable	D_InOPREV		D_InIPREV			D_ROA		D_PRODUC		D_OCC
AI_{it}	0.083*** (0.024)		0.122*** (0.040)			0.091 (0.072)		0.004** (0.002)		0.020** (0.009)
Control variables	Yes		Yes			Yes		Yes		Yes
Year and State	Yes		Yes			Yes		Yes		Yes
Fixed effects										
R^2	0.315		0.187			0.202		0.200		0.240
Number of observations	420		420			421		419		421
Periods: –1 to +3										
Variable	D_InOPREV		D_InIPREV			D_ROA		D_PRODUC		D_OCC
AI_{it}	0.085*** (0.032)		0.139** (0.057)			0.155 (0.150)		0.006** (0.002)		0.029** (0.011)
Control variables	Yes		Yes			Yes		Yes		Yes
Year and State	Yes		Yes			Yes		Yes		Yes
Fixed effects										
R^2	0.511		0.266			0.278		0.227		0.292
Number of observations	285		286			286		284		286

This table employs a difference-in-difference test to identify the causality between AI adoption and performance. Sample sizes vary across models because some hospitals do not have data for all performance categories. Robust standard errors are reported in parentheses.

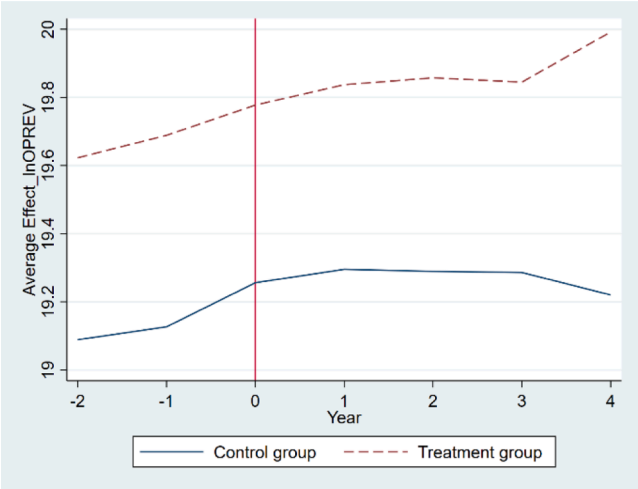
*, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

uptick. Another pertinent aspect lies in the composition of our sample. A substantial portion, approximately 75 %, consists of nonprofit and government hospitals. These entities may prioritize objectives such as offering charitable care or enhancing the quality of treatment over the maximization of net income. Consequently, the utilization of ROA as a metric for such nonprofit and government hospitals might not yield results in alignment with conventional expectations.

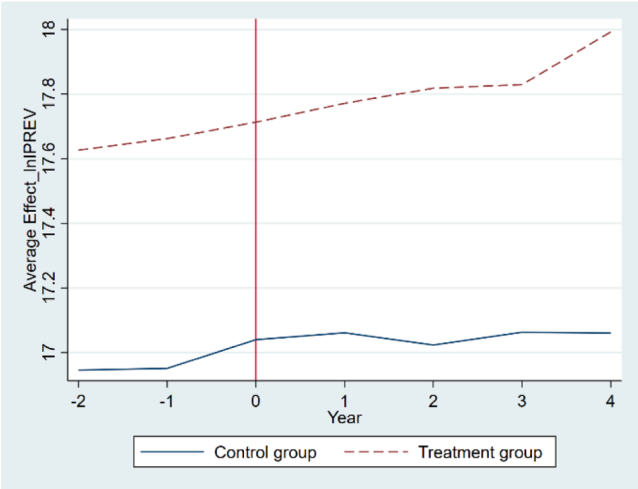
6.2. Limitations and future research

As in any research, this study is not without limitations. First, this study exclusively employs an indicator variable to measure the status of AI adoption. This simplified approach may overlook variations in AI techniques utilized and the extent of AI adoption. The diversity could entail differences between adopting AI across an entire hospital versus specific departments, or the frequency of AI usage (e.g., daily adoption

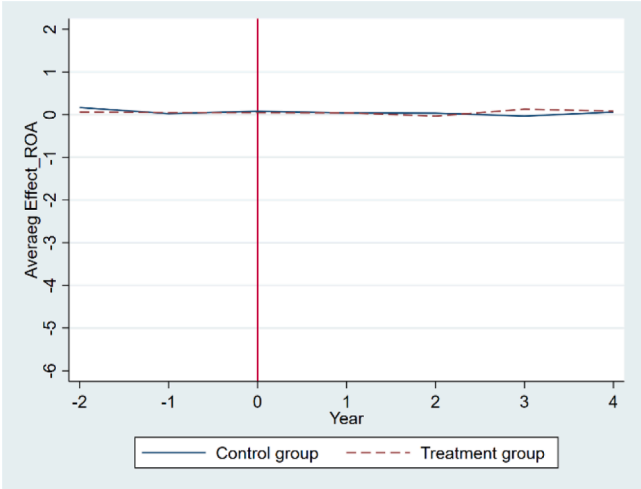
vs. as-needed). To comprehensively capture these nuances, future investigations might supplement secondary data with survey-based or field-based data, thereby enabling a more accurate assessment of AI types and degrees of adoption. Second, due to data limitations, we have been unable to include several significant control variables in our analysis. These variables, including factors such as the number of doctors, doctor capabilities, the count of IT artifacts or systems used in the hospitals, and the extent of specialties or services offered by the hospital, are essential for a comprehensive examination. We recognize the importance of these variables and recommend that future research endeavors consider their incorporation to enrich the analytical framework. Third, while we addressed endogeneity concerns through Wooldridge's instrumental variable approach, the possibility of self-selection bias cannot be entirely disregarded (Maiga, 2014). This intriguing aspect warrants further exploration, thus signifying an area ripe for future research. Fourth, our study omits an examination of the performance



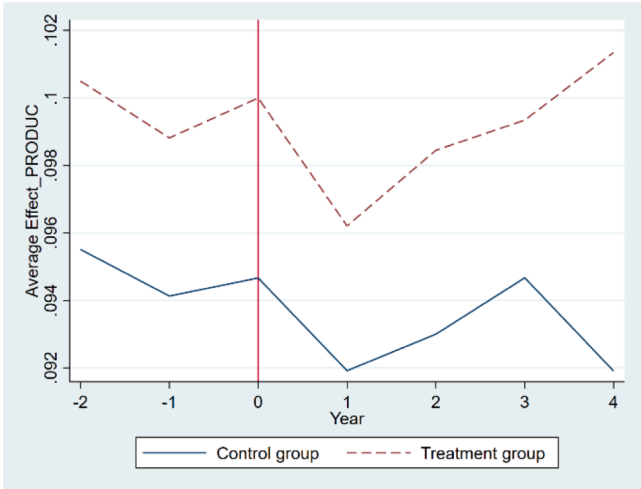
(a) lnOPREV



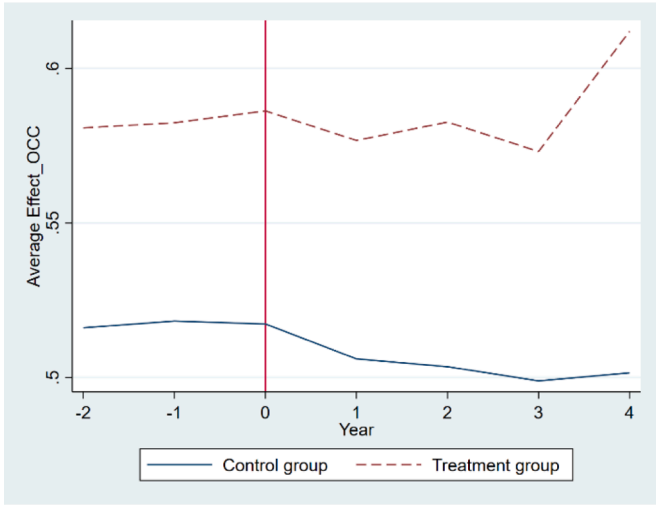
(b) lnIPREV



(c) ROA



(d) PRODUC



(e) OCC

Fig. 2. Evolution of Performance in Control and Treatment Groups.

Table 8
Traditional 2SLS Analysis.

Variable	lnOPREV _{it}	lnIPREV _{it}	ROA _{it}	PRODUC _{it}	OCC _{it}
<i>Intercept</i>	12.614*** (0.322)	8.410*** (0.774)	−2.261*** (1.168)	−0.008 (0.028)	−0.581*** (0.132)
<i>FittedProbability_{it}</i>	4.835*** (0.163)	6.189*** (0.395)	−0.259 (0.237)	0.064*** (0.014)	0.424*** (0.056)
<i>CMI_{it}</i>	0.420*** (0.043)	0.237** (0.103)	−0.220*** (0.180)	−0.025*** (0.004)	0.015 (0.015)
<i>Size_{it}</i>	0.190*** (0.017)	0.281*** (0.042)	0.138* (0.071)	0.003 (0.001)	0.050*** (0.006)
<i>Medicare%_{it}</i>	−0.321*** (0.114)	−0.337 (0.270)	−0.054 (0.210)	−0.026** (0.011)	−0.047 (0.046)
<i>Medicaid%_{it}</i>	1.216*** (0.171)	3.447*** (0.358)	−0.152 (0.165)	0.045*** (0.015)	0.187*** (0.059)
<i>Teaching_{it}</i>	0.011 (0.027)	0.100* (0.055)	−0.028 (0.022)	0.003 (0.002)	0.045*** (0.010)
<i>Metro_{it}</i>	0.188*** (0.034)	0.424*** (0.067)	−0.019 (0.045)	0.016 (0.003)	0.032** (0.012)
<i>FP_{it}</i>	0.571*** (0.044)	0.754*** (0.092)	0.225*** (0.079)	0.037*** (0.004)	0.030** (0.015)
<i>NP_{it}</i>	0.273*** (0.038)	0.417*** (0.080)	0.063 (0.052)	0.013*** (0.003)	0.034*** (0.012)
<i>System_{it}</i>	−1.350*** (0.066)	−1.600*** (0.159)	0.082 (0.097)	−0.002*** (0.006)	−0.074*** (0.022)
<i>State</i>	Included	Included	Included	Included	Included
<i>Year</i>	Included	Included	Included	Included	Included
<i>R²</i>	0.889	0.790	0.079	0.454	0.635
Number of observations	1,480	1,482	1,483	1,481	1,483

This table employs the traditional 2SLS analysis to test the robustness of our results. Sample sizes vary across models because some hospitals do not have data for all performance categories. Robust standard errors are reported in parentheses. See Appendix A for variable definitions.

*, **, *** Indicate statistical significance at $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively, in a two-tailed t -test.

differentials between early and late adopters. As elucidated by [Jacobs et al. \(2015\)](#), the initial stages of diffusion might engender challenges due to limited experience or knowledge to guide adoptions, potentially rendering administrative innovations more intricate. Delving into the distinctive performance outcomes of early versus late AI adopters stands as an intriguing prospect for future research. This would involve the evaluation of performance outcomes by comparing different subgroups based on their respective timings of AI adoption.

In closing, while acknowledging these limitations, we hope that our research provides valuable insights for practitioners and serves as a foundation for further investigations within the realm of AI in healthcare.

CRediT authorship contribution statement

Phuoc Pham: Conceptualization, Data curation, Writing – original draft, Writing – review & editing. **Huilan Zhang:** Writing – review &

editing, Writing – original draft, Methodology, Formal analysis. **Wenlian Gao:** Writing – review & editing, Methodology. **Linda Xiaowei Zhu:** Writing – original draft, Writing - review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Variable definitions

Variables	Definition
Determinants	
<i>MKTSHR</i>	The number of patient days for each hospital / the total number of patient days of hospitals within a radius of 25 miles
<i>ALOS</i>	Total number of inpatient days / the total number of inpatients discharged
Performance Measures	
<i>lnOPREV</i>	Natural logarithm of total outpatient revenue
<i>lnIPREV</i>	Natural logarithm of total inpatient revenue
<i>ROA</i>	Net income / Total assets at year-end
<i>PRODUC</i>	Total number of patients per year / (total number of full-time employees × 365)
<i>OCC</i>	Total number of patients per year / (total number of licensed beds × 365)
Control Variables	
<i>lnTotalExp</i>	Natural logarithm of total expenses
<i>CMI</i>	Case Mix Index, measuring service complexity
<i>Size</i>	Natural logarithm of total assets at year-end
<i>Medicare%</i>	Medicare patient days / Hospital patient days
<i>Medicaid%</i>	Medicaid patient days / Hospital patient days

(continued on next page)

(continued)

Variables	Definition
Teaching	An indicator variable that takes the value of one for teaching hospitals and zero otherwise
Metro	An indicator variable that takes the value of one for hospitals located in metro areas and zero otherwise
FP	An indicator variable that takes the value of one for for-profit hospitals and zero otherwise
NP	An indicator variable that takes the value of one for nonprofit hospitals and zero otherwise
System	An indicator variable that takes the value of one for system-affiliated hospitals and zero otherwise

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